Horticulture 1 · Unit 3 · Level 2 (modified)

Parts of the Plant



TODAY'S PLAN

- 🌱 Roots
- 🎋 Stems: xylem and phloem
- 🍃 Leaves and photosynthesis
- 🌸 Flowers and pollination

Review · 5 minutes

Quick Review



🏷️ Write a scientific name: what is capitalized?

🌼 Name the 3 flower types.

⏳ Annual, biennial, perennial: give one example.

Words to Know

Words to Know



Root

Grows underground, anchors the plant, and takes in water and minerals through root hairs.



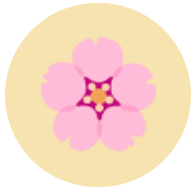
Stem

Holds up leaves and flowers and carries water and food through xylem and phloem.



Leaf

The main job is photosynthesis: making food (sugar) from light, water and carbon dioxide.



Flower

The reproductive part. It makes seeds and fruit after pollination.

Words to Know

More Words to Know



☀️ **Photosynthesis**: Water + carbon dioxide + light → sugar (food) + oxygen.

🐝 **Pollination**: Moving pollen from the stamen to the pistil, often by bees. It leads to seeds and fruit.

💧 **Xylem**: Tubes in the stem that carry water and minerals up from the roots.

🍬 **Phloem**: Tubes that carry food (sugar) through the plant.

Compare

Roots



Taproot

One thick main root

carrot, dandelion

Fibrous roots

Many thin roots

grasses, corn

Learn

What Roots Do



Anchor the plant and **absorb** water and minerals through **root hairs**.

Store food (carrots, beets).

The **root cap** protects the growing tip (it does NOT absorb water).

🃏 Check 1 of 3



Which part absorbs most water?

Hold up 1, 2, or 3:

1️⃣ Root cap

2️⃣ Root hairs

3️⃣ Petals

✅ Answer · Card 1 of 3



Root hairs

Root hairs absorb; the cap protects.

Learn

Stems

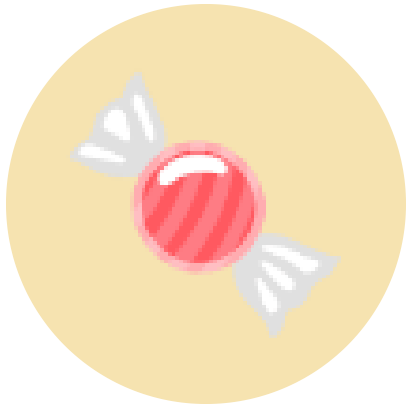


Xylem carries water and minerals **up**.

Phloem carries **food** through the plant.

Dicots: rings with cambium; monocots: scattered **vascular bundles**.

🃏 Check 2 of 3



Phloem carries…

Hold up 1, 2, or 3:

1️⃣ Only water upward

2️⃣ Pollen

3️⃣ Food (sugar)

✅ Answer · Card 2 of 3



Food (sugar)

Xylem = water up; phloem = food.

Learn

Leaves



Main job: **photosynthesis** (food).

Petiole = the stalk that attaches the leaf.

Stomata open and close for gas exchange and **transpiration** (giving off water).

Learn

Describing Leaves



Form

needle, round, ovate



Margin

edge: dentate, undulate



Arrangement

opposite, alternate, whorled

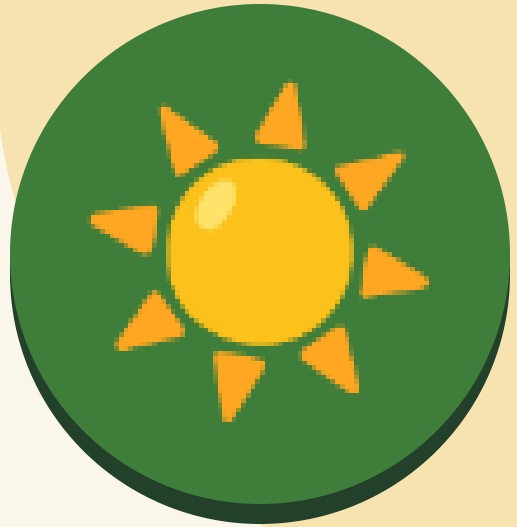


Growth habit

upright, vine, cascading

Learn

Photosynthesis

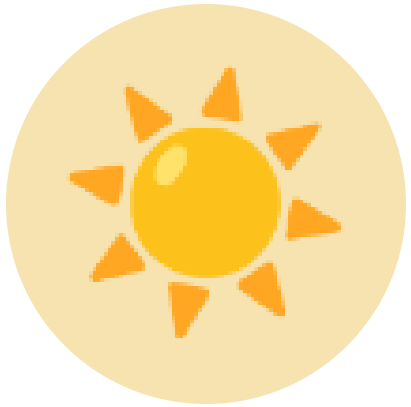


In: water + carbon dioxide (+ light).

Out: sugar (food) + oxygen.

It happens in the green parts of leaves.

🃏 Check 3 of 3



The products of photosynthesis are…

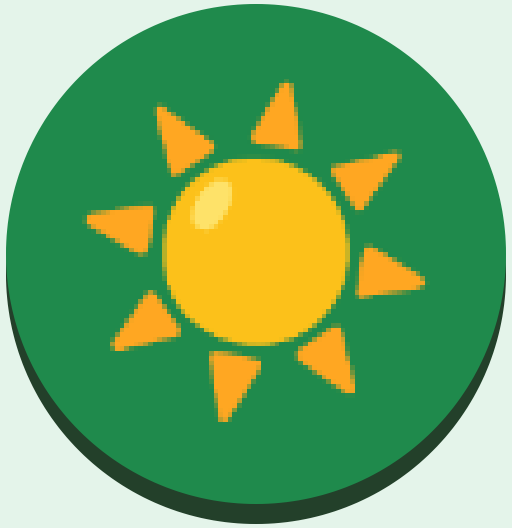
Hold up 1, 2, or 3:

1️⃣ Oxygen and food

2️⃣ Water and carbon dioxide

3️⃣ Soil and sun

✅ Answer · Card 3 of 3

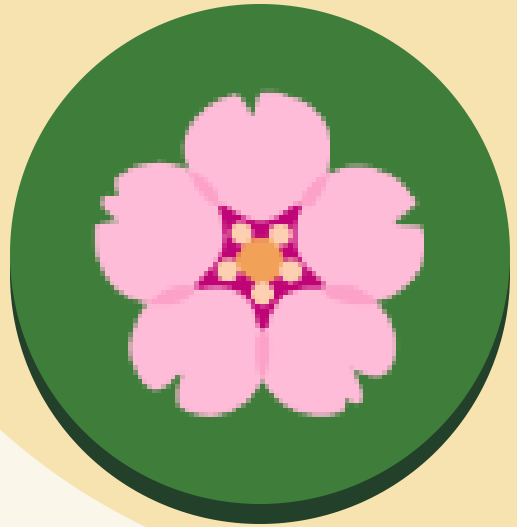


Oxygen and food

Inputs: water + CO2.

Learn

Flowers and Pollination



Stamen (male) = anther + filament, makes pollen.

Pistil (female) = stigma, style, ovary, ovules.

Pollination → **seed and fruit**. Bees are key pollinators.

Wrap-Up

Recap



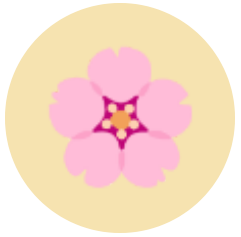
Roots
anchor + absorb



Xylem
water up



Phloem
food



Pollination
seed + fruit

Wrap-Up

Exit Ticket



Xylem vs. phloem: what does each carry?

Name the inputs and outputs of photosynthesis.

Which flower part makes pollen?